MILITARY SPECIFICATION

MASK, SURGICAL, DISPOSABLE

This Amendment forms a part of Military Specification MIL-P-36431B, dated 27 January 1970, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 4:

3.1.2.1 Facing material. Delete first sentence entirely and substitute: "The facing material shall be a non-woven fabric of suitable fibers."

3.1.2.2 Filter media. At end of first sentence, delete period and add: "and/or other suitable fibers." Delete the second sentence entirely.

Page 11:

4.5.2.1.1 Delete entirely and substitute:

"4.5.2.1.1 Method 1. Cultures of Staphylococcus aureus, Type number 15 (see 6.4) cultures are transferred at least 3 times in nutrient broth (BBL) before preparation of the test suspension. Each transfer is incubated at 37° ± 2°C. for 24 to 28 hours, constantly agitated throughout the incubation period on a mechanical shaker.

Aerosol filtration of bacteria. The efficiency of each mask is tested with the apparatus shown in Figures I and II in which an Andersen sample (see Fig. I) is utilized to collect and count bacteria on nutrient agar (BBL) in special petri dishes.

The bacterial culture is first diluted to a given turbidity as may be measured by an adequate optical instrument (i.e. Klett-Summerson or Coleman Jr. spectrophotometers) such that when the optically adjusted suspension is diluted approximately 1/3500, the nutrient agar control plates will yield a total colony count of 2200 ± 500 colonies. The diluted culture suspension is fed into a nebulizer from a syringe driven by a constant drive mechanism (see figure 2). The nebulizer is operated at 15 pounds per square inch pressure to produce an aerosol in the aerosol chamber. The aerosol is drawn through the apparatus at a flow rate of 28.3 liters per minute for one minute to establish control values. The air flow is measured with a Roger Gilmont Instruments (RGI) flowmeter or other flowmeter of comparable precision. The nebulizer is then turned off and the chamber and sampler washed with 28.3 liters of air. The control plates are removed from the sampler and replaced immediately with fresh nutrient agar plates. Thereafter, circular specimens, 3 inches in diameter, of each surgical mask to be tested are interposed, in turn, at the point where the aerosol chamber is attached to the Andersen sampler. The bacterial aerosol is again drawn through the apparatus, identical to the control run, except for the presence of the mask material. All plates are incubated at 37° ± 2°C. for 48 hours and the colonies counted.

6515

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The filtration efficiency is reported in "Percent Efficiency"

$$\text{Percent Efficiency} = \frac{C-F}{C} \times 100$$

Where, C = the average number of colonies on control plates.
F = number of colonies on test plates (mask material interposed)."

Page 15:

5.2.1.1 Exterior container. Delete and substitute:

"5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. The special requirements as specified in Table II of PPP-B-636 shall apply for fiberboard boxes. Closure and strapping shall be as specified in the appendix of the applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution."

5.2.1.2 Case liner. At end of paragraph add: "Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636."

Page 16:

5.2.2.1 Exterior container. Delete and substitute:

"5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. The special requirements as specified in Table I of PPP-B-636 shall apply for fiberboard boxes. Closure of fiberboard boxes shall conform to method II of PPP-B-636. All other boxes shall be closed as specified in the appendix of the applicable box specification."

Add new paragraph, as follows:

"6.4 Cultures of Staphylococcus aureus, Utah strain 15, are available from:

Dr. Paul S. Nicholes
Dept. of Microbiology
University of Utah Medical Center
Salt Lake City, Utah 84112"

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Custodians:
Army - MD
Navy - MS
Air Force - 03

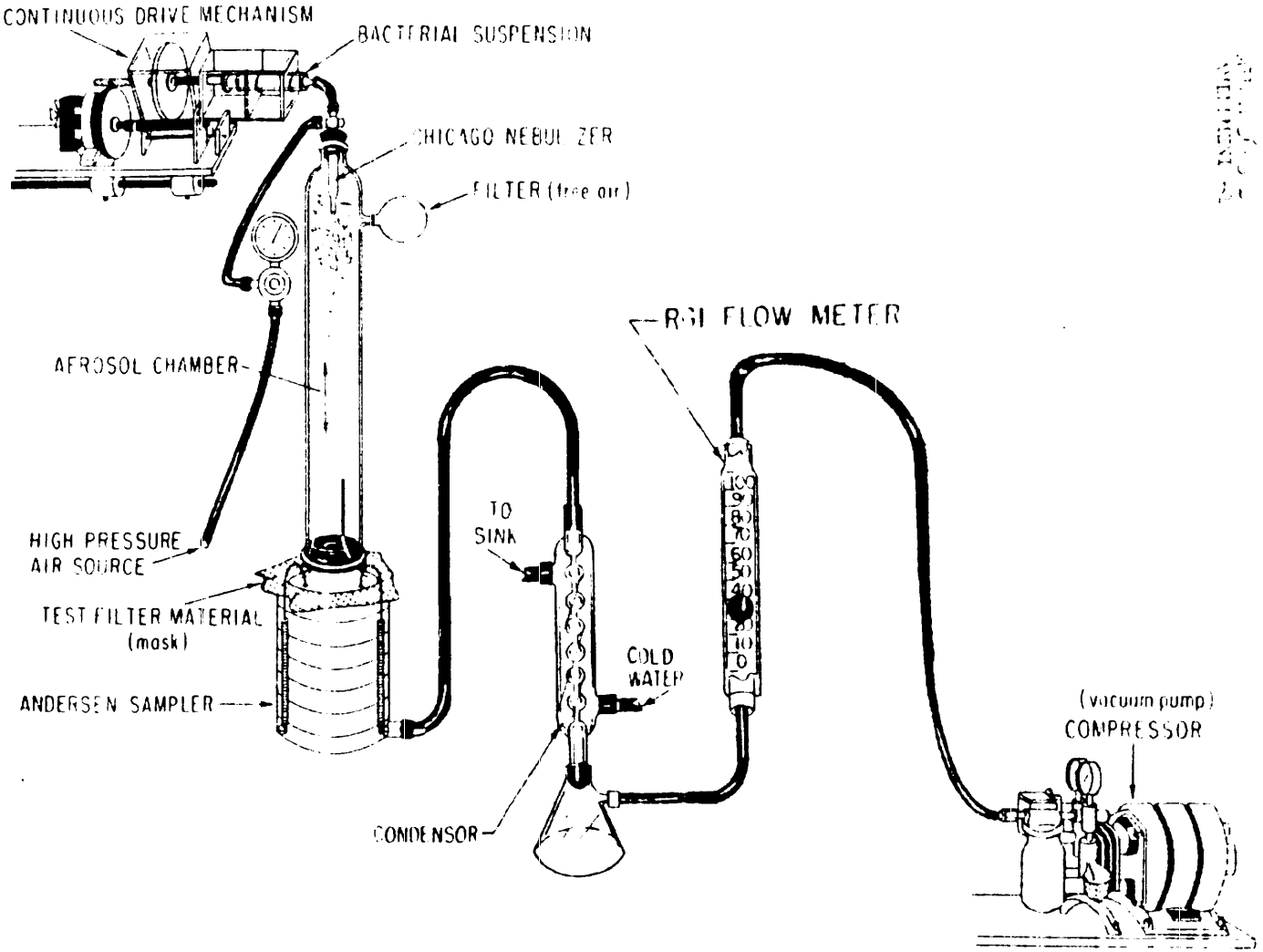
Review activities:
Army - MD
Navy - MS
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Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1675

Review information is current as of the date of this document. For future coordination, draft circulation should be based on information in the current DODISS.

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FIGURE 1

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